

An Efficient Synthesis of Diaryl Ketones by Iron-Catalyzed Arylation of Aryl Cyanides**

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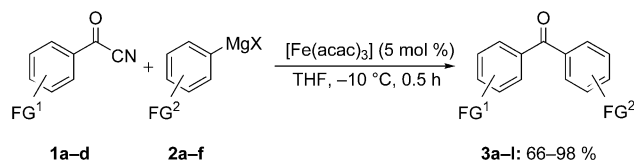
Dedicated to Professor Klaus T. Wanner on the occasion of his 50th birthday

The acylation of organometallic intermediates with acid chlorides is an important method for the preparation of polyfunctional ketones. This functionality is present in a great variety of pharmaceutical and material-science target molecules.^[1] Many organometallic reagents have been used for performing acylations, and organomanganese reagents have proved to be especially useful.^[2] Polyfunctional organozinc compounds have also been used frequently, and smooth acylations can be performed in the presence of stoichiometric amounts of CuCN·2LiCl^[3] or in the presence of a palladium catalyst.^[4] The preparation of functionalized arylzinc reagents is less straightforward^[5] and requires cobalt catalysis^[6] or the use of activated zinc powder (Rieke zinc).^[7]

Recently we reported a general method for preparation of polyfunctional arylmagnesium halides of type **2** using I/Mg- or Br/Mg-exchange reactions.^[8] We envisioned using these organometallics for the preparation of polyfunctionalized diaryl ketones by their reaction with acid chlorides. In preliminary experiments we treated benzoyl chloride with PhMgCl at various temperatures and obtained yields between 50–58%. We then turned our attention towards [Fe(acac)₃]-catalyzed reactions.^[9,10] The reaction of PhMgCl with PhCOCl in the presence of [Fe(acac)₃] (5 mol %) at 0 °C or 20 °C afforded benzophenone in only 38–53% yield. Extensive variation of the experimental reaction conditions (concentration, addition time, inverse addition) did not improve these results. This is in agreement with literature reports

which indicate that the preparation of benzophenone derivatives by the acylation of organometallics is only a moderately efficient reaction.^[1a]

We then examined the use of acyl cyanides **1** as the acylating agents^[11,12] and found that they readily react with various aromatic organomagnesium compounds of type **2** leading to polyfunctional benzophenone derivatives of type **3** (Scheme 1 and Table 1). Acyl cyanides are more powerful



Scheme 1. [Fe(acac)₃]-catalyzed reactions of functionalized magnesium reagents with acyl cyanides. For the functional groups (FG), see Table 1.

acylating agents than acid chlorides, since the cyano group enhances the reactivity of the adjacent carbonyl group. In contrary, the chlorine atom of acid chlorides plays the role of a donor by a mesomeric effect. The reaction of benzoyl cyanide (**1a**) and phenylmagnesium chloride (**2a**) without the iron catalyst furnished a higher yield than that obtained for the addition of phenylmagnesium chloride (**2a**) to benzoyl chloride (75% vs. 58% at 0 °C).

However, the use of catalytic amounts of [Fe(acac)₃] (5 mol %) was beneficial for the reaction of 4-ethoxycarbonylphenylmagnesium chloride (**2b**) with benzoyl cyanide (**1a**), increasing the yield from 58% to 80% at -10 °C.^[13] Thus, the reaction of PhMgCl (**2a**) with PhCOCN (**1a**) provided benzophenone (**3a**) in 84% yield (entry 1 of Table 1). Similarly, functionalized organomagnesium compounds **2b** and **2c** reacted in the presence of [Fe(acac)₃] (5 mol %) with PhCOCN (**1a**) at -10 °C within 0.5 h, furnishing the expected benzophenones **3b** and **3c** in 80 and 78% yield, respectively (entries 2 and 3). Functionalized acyl cyanides bearing a chlorine (**1b**), a methoxy (**1c**), or a ethoxycarbonyl group (**1d**) in *para* position, (entries 4–12) reacted with various arylmagnesium reagents (**2b–f**), leading to the diaryl ketones **3d–l** in good yields. Interestingly, an *ortho*-substituted arylmagnesium species like **2e** reacted as well, furnishing the benzophenone **3i** in 66% yield (entry 9). Also ketones bearing heterocyclic groups were prepared. The reaction of acyl cyanide **1c** with the heterocyclic Grignard reagent **2f** led to furyl ketone **3l** in 78% yield (entry 12). Heterocyclic acyl cyanides, like pyridine derivative **4**, reacted under our standard conditions (-10 °C, 0.5 h) with various aryl magnesium reagents, such as **2a**, **2b**, and **2d**, to give pyridyl ketones **5a–c** in 75–86% yields (Scheme 2).

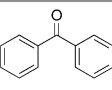
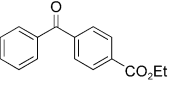
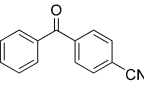
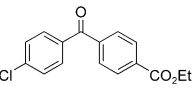
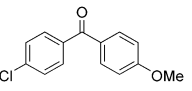
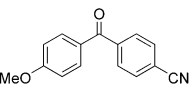
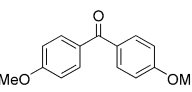
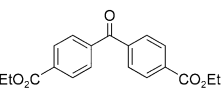
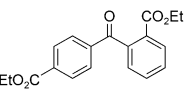
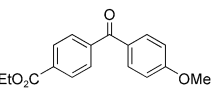
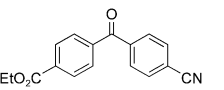
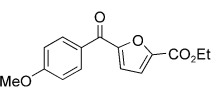
In summary, we have shown that the arylation of aryl and heteroaryl acyl cyanides with functionalized aryl and heteroaryl magnesium species is efficiently catalyzed by [Fe(acac)₃] (5 mol %), furnishing a range of new polyfunctional diaryl ketones.

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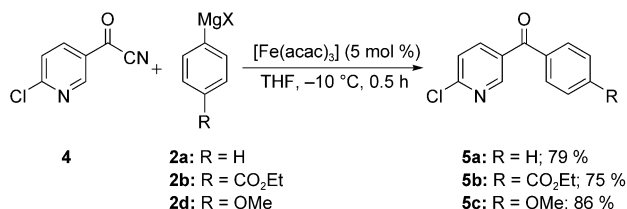
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Supporting information for this article is available on the WWW under <http://www.angewandte.org> or from the author.

Table 1: Ketones **3a–l** obtained by the Fe^{III} -catalyzed reaction of aryl acyl cyanides **1a–d** with aryl magnesium halides **2a–f** (Scheme 1).

Entry	Acyl cyanide	Grignard reagent	Product	Yield [%] ^[a]
1				84
2				80
3				78
4				74
5				89
6				84
7				98
8				68
9				66
10				83
11				71
12				78

[a] Yield of isolated, analytically pure product.



Scheme 2. $[\text{Fe}(\text{acac})_3]$ -catalyzed reactions of heterocyclic pyridyl acyl cyanide **4**.

Experimental Section

Typical procedure for the arylation of aryl cyanides.

3k (entry 11, Table 1): A dry and argon-flushed 10-mL flask equipped with a stirring bar and a rubber septum was charged with anhydrous THF (5 mL) and 4-iodobenzonitrile (374 mg, 2.4 mmol). The solution was cooled to -20°C and isopropylmagnesium chloride (1.9 mL, 1.4 M in THF, 2.6 mmol) was added slowly. The reaction mixture was stirred at this temperature until the exchange reaction was complete (30 min, checked by GC analysis of reaction aliquots). The resulting solution was then transferred dropwise over 25 min by cannula into a second 50-mL flask, which contained a solution of **1d** (406 mg, 2.0 mmol) and $[\text{Fe}(\text{acac})_3]$ (35 mg, 0.1 mmol) in anhydrous THF (10 mL) stirred at -10°C . At the end of the addition, the reaction mixture was quenched with aq. NH_4Cl (10 mL), diluted with water (25 mL), and extracted with Et_2O (3×25 mL). The combined organic layers were washed with aq. NaHCO_3 (10 mL), brine (2×20 mL), and dried (MgSO_4) and were concentrated in vacuo. The residue was purified by flash chromatography on silica gel (pentane/diethyl ether 4:1) yielding the diaryl ketone **3k** (397 mg, 71 %) as a colorless solid (m.p. $110\text{--}112^\circ\text{C}$).

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Keywords: acyl cyanides · homogeneous catalysis · iron · ketones · organomagnesium reagents

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- [12] Aryl acyl cyanides are best prepared by the reaction of the corresponding acyl chlorides with copper cyanide in refluxing acetonitrile (ref. [11 b]).
- [13] Lower reaction temperatures like -20 , -40 , and -80°C increased the yield only slightly to 65, 70, and 68 %, respectively.